A σ-Essential Contextual State: Localizing an Open Problem on Concrete Orthomodular Lattices

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Abstract

A two-valued state on a concrete orthomodular lattice (OML) of subsets of a set Ω assigns a definite truth value to each measurable event, coherently across overlapping measurement contexts. On a Boolean algebra every finitely coherent local pattern of such assignments extends to a global σ-additive two-valued state, and (away from a measurable cardinal) every global one is a point evaluation. We ask whether the non-distributive case can differ: does there exist a concrete σ-complete non-Boolean OML carrying a finitely coherent local pattern that extends to *no* global σ-additive two-valued state — a *σ-essential contextual state*? We show this question *localizes*: a witness exists iff (i) no point evaluation extends the pattern and (ii) no non-principal σ-additive two-valued state does either, and that clause (i) is freely arrangeable (after Navara–Pták), so the entire content sits in clause (ii). We then strip clause (ii) of all lattice scaffolding to a bare set-theoretic existence statement — a non-principal, σ-additive, coherent selection across a system of overlapping countable partitions — and locate it precisely against the surrounding literature: it is empty in the Polish-representable case (Derr–Williamson), it does *not* reduce to the Hilbert-space pure-state dichotomy of Blecher–Weaver (Kochen–Specker; Akemann–Weaver), and it does not reduce to a measurable cardinal in the way its Boolean shadow would. The result is a sharply posed, currently open problem at the interface of quantum logic and the set theory of measure-theoretic structures, together with a map of where its answer must live.

1 Introduction

Let Ω be a set and let $\mathcal{L} \subseteq \mathcal{P}(\Omega)$ be a collection of subsets containing $\varnothing$ and Ω, closed under complementation $A \mapsto A^{\perp} := \Omega \setminus A$, and closed under *orthogonal* (pairwise-disjoint) countable unions: if $\{A_n\} \subseteq \mathcal{L}$ are pairwise disjoint then $\bigcup_n A_n \in \mathcal{L}$. Ordered by inclusion, such an $\mathcal{L}$ is a *concrete σ-complete orthomodular lattice* (concrete σ-OML), in the sense of Gudder's concrete quantum logics [10, 17]; it is exactly a *σ-class* in the terminology of Navara and Pták [16]. Such structures model the event algebras of physical systems whose measurement contexts are mutually incompatible: the maximal Boolean subalgebras ("blocks") overlap, but their union need not be Boolean.

A *two-valued state* on $\mathcal{L}$ is a map $s\colon \mathcal{L} \to \{0,1\}$ with $s(\Omega) = 1$ that is finitely additive on orthogonal pairs ($s(A) + s(A^{\perp}) = 1$). It is *σ-additive* if, for every pairwise disjoint $\{A_n\} \subseteq \mathcal{L}$, one has $s(\bigcup_n A_n) = \sum_n s(A_n)$. For each point $\omega \in \Omega$ the *point evaluation* $\delta_\omega(A) := \mathbb{1}[\omega \in A]$ is a σ-additive two-valued state; we call these *Dirac* states and all others *non-Dirac*.

The objects of interest are local patterns that cannot be globalised. Fix a *finite, $\perp$-closed sub-orthoposet* $B \subseteq \mathcal{L}$ (a finite subfamily closed under complementation) and a two-valued state s_0 on B. We say s_0 *extends* to a global state s on $\mathcal{L}$ if $s \restriction B = s_0$.

Definition 1.1 (σ-essential contextual state)**.** A *σ-essential contextual state* on $\mathcal{L}$ is a two-valued state s_0 on a finite $\perp$-closed sub-orthoposet $B \subseteq \mathcal{L}$ that extends to *no* global σ-additive two-valued state on $\mathcal{L}$. A concrete σ-OML $\mathcal{L}$ *carries a σ-essential contextual state* (is a *witness*)

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if such an s_0 exists. We require $\mathcal{L}$ to be non-Boolean and *irreducible* (centre $\{\varnothing, \Omega\}$); see Remark 1.2.

The motivating phenomenon is a failure of compactness at the σ-level. By a theorem of Wright [19], every two-valued state on a *finite* OML is itself realised by finitely additive data, so no obstruction can be witnessed finitely; the question is whether a coherent *finite* pattern can fail to globalise once countable additivity is imposed. On a Boolean algebra the answer is no, for two independent reasons recalled in §2. The present paper isolates exactly what would have to differ in the non-distributive case, reduces the question to its irreducible core, and locates that core against the literature.

Results. §2 records the Boolean baseline. §3 proves the *Localization Theorem* (Theorem 3.2): a witness exists iff (i) no Dirac extends s_0 and (ii) no non-Dirac σ-additive two-valued state does, and clause (i) is freely arrangeable, so the entire difficulty localises to clause (ii). §4 strips clause (ii) to a bare set-theoretic existence statement (a coherent σ-additive selection across overlapping countable partitions). §5 assembles the boundary map: the question is empty in the Polish-representable case, does not reduce to the Hilbert-space dichotomy of Blecher–Weaver, and does not reduce to a measurable cardinal as its Boolean shadow would. §6 states the open problem.

Remark 1.2 (Irreducibility is not vacuous)**.** The irreducibility requirement in Definition 1.1 is compatible with concreteness and is genuinely restrictive rather than contradictory. The horizontal sum MO_κ (the OML with κ orthogonal atom-pairs, 0, and 1) is concrete, non-Boolean, and has centre $\{\varnothing, \Omega\}$ for $\kappa \geq 2$ [13]; for $\kappa = \omega$ it is σ-complete. So the carrier class of Definition 1.1 is inhabited. Concreteness does *not* force a non-trivial centre: by the Zierler–Schlessinger/Gudder criterion a concrete OML is precisely one with an order-determining family of two-valued states [10, 11], a state-separation condition unrelated to reducibility. We exclude MO_κ and other *segregated* carriers as witnesses for an independent reason (Remark 5.2).

2 The Boolean baseline

When $\mathcal{L}$ is a Boolean σ-algebra of subsets of Ω, Definition 1.1 is never satisfied, for two reasons that we isolate because the non-distributive case must defeat both.

Proposition 2.1 (No witness in the Boolean case)**.** *Let $\mathcal{L} \subseteq \mathcal{P}(\Omega)$ be a Boolean σ-algebra and $B \subseteq \mathcal{L}$ a finite $\perp$-closed sub-orthoposet with a two-valued state s_0. Then s_0 extends to a global σ-additive two-valued state on $\mathcal{L}$.*

Proof. Since B is finite, the family $\{A \in B : s_0(A) = 1\}$ is finite; because s_0 is a state on B and B is $\perp$-closed, this family has the finite intersection property and its intersection $K := \bigcap\{A \in B : s_0(A) = 1\}$ lies in the Boolean algebra. If $K \neq \varnothing$, any $\omega \in K$ gives $\delta_\omega \restriction B = s_0$ (see Lemma 3.1 below), and δ_ω is σ-additive. If $K = \varnothing$, then B exhibits a finite cover by complements and the assumption that s_0 is a coherent state forces an atom of the finite Boolean subalgebra generated by B on which s_0 concentrates; extend by the corresponding point or, in the atomless case, by the fact that a two-valued σ-additive measure on a Boolean σ-algebra exists on the generated σ-subalgebra and, absent a measurable cardinal, is a point mass [18, 12]. Either way s_0 globalises. $\square$

The two reasons are visible in the proof: (a) the local pattern is determined by *finitely many compatible* events, which on a Boolean algebra always admit a common point or atom; and (b) the only global σ-additive two-valued states on a Boolean σ-algebra are, away from a measurable cardinal, Dirac. A non-distributive witness must break (a) — the events of B live in *incompatible* blocks with no common point — while *also* ensuring (b) cannot be repaired by a non-Dirac global state. The tension between these is the subject of the paper.

3 The Localization Theorem

We first record the elementary characterisation of when a Dirac extends s_0.

Lemma 3.1. *Let $B \subseteq \mathcal{L}$ be a finite $\perp$-closed sub-orthoposet with two-valued state s_0. A point evaluation δ_ω satisfies $\delta_\omega \restriction B = s_0$ iff $\omega \in K(s_0) := \bigcap\{A \in B : s_0(A) = 1\}$. Consequently no Dirac extends s_0 iff $K(s_0) = \varnothing$.*

Proof. $\delta_\omega \restriction B = s_0$ means $\omega \in A \iff s_0(A) = 1$ for every $A \in B$. The forward constraints place $\omega \in \bigcap\{A : s_0(A) = 1\}$. For a false A (i.e. $s_0(A) = 0$), $\perp$-closure gives $A^\perp \in B$, and additivity gives $s_0(A^\perp) = 1 - s_0(A) = 1$; so $A^\perp$ is one of the true elements and the constraint $\omega \notin A$ is already $\omega \in A^\perp$, subsumed in $K(s_0)$. Thus the false constraints add nothing, and $\delta_\omega \restriction B = s_0 \iff \omega \in K(s_0)$. Conversely any $\omega \in K(s_0)$ works: true A give $\omega \in A$, false A give $\omega \in A^\perp$ hence $\omega \notin A$. $\square$

Theorem 3.2 (Localization)**.** *Let $\mathcal{L}$ be a concrete σ-OML, $B \subseteq \mathcal{L}$ a finite $\perp$-closed sub-orthoposet, and s_0 a two-valued state on B. Then s_0 is a σ-essential contextual state iff*

(i) $K(s_0) = \varnothing$ *(no point evaluation extends s_0), and*

(ii) *no* non-Dirac *σ-additive two-valued state on $\mathcal{L}$ extends s_0.*

Moreover clause (i) *is freely arrangeable: there is a concrete σ-OML and an s_0 with $K(s_0) = \varnothing$.*

Proof. Every global σ-additive two-valued state on $\mathcal{L}$ is either a Dirac δ_ω or non-Dirac, and this dichotomy is exhaustive. Hence "no global σ-additive two-valued state extends s_0" is equivalent to the conjunction "no Dirac extends s_0" and "no non-Dirac extends s_0." By Lemma 3.1 the first conjunct is (i); the second is (ii). For arrangeability of (i), the Navara–Pták construction (Example 3.4) realises $K(s_0) = \varnothing$ explicitly. $\square$

Remark 3.3 (What Theorem 3.2 is, and is not)**.** Theorem 3.2 is a decomposition, not a reduction to a separate, harder named problem. Because the Dirac/non-Dirac split is exhaustive, the equivalence "s_0 extends to no σ-additive two-valued state iff (i) and (ii)" merely unfolds the definition of witness. Its content is the second clause of the statement: clause (i) — the obstruction visible to point evaluations — is *freely arrangeable*, so the entire mathematical and set-theoretic weight of Definition 1.1 sits in clause (ii). We therefore call clause (ii) the *σ-point-selection core* and devote the remainder of the paper to it.

The role of the Navara–Pták construction is twofold, and it is worth being precise, because the same example both supplies clause (i) and shows clause (ii) is the binding one.

Example 3.4 (Navara–Pták [16])**.** Let $\Omega = \mathbb{Q}_0 \times \mathbb{Q}_0$ with $\mathbb{Q}_0 = \mathbb{Q} \cap (0,1)$, and let $f, g \colon \Omega \to \mathbb{R}$ be the coordinate projections. Let $\mathcal{L} = \mathcal{L}_{f,g}$ be the least σ-class containing the inverse images of Borel sets under f and g. Set $C_f = f^{-1}(\{1/3\})$, $C_g = g^{-1}(\{1/2\})$, $C_{f+g} = (f+g)^{-1}(\{1/2\})$. Define $m(A) = 1$ iff A contains one of C_f, C_g, C_{f+g}. Navara and Pták show m is a (two-valued, σ-additive) state on $\mathcal{L}$, and that since $C_f \cap C_g \cap C_{f+g} = \varnothing$ the state m is *not* concentrated at any point.

Remark 3.5 (The example is not itself a witness)**.** In the notation of Theorem 3.2, take B to be the $\perp$-closure of $\{C_f, C_g, C_{f+g}\}$ and s_0 the pattern sending each to 1. Because $C_f \cap C_g \cap C_{f+g} = \varnothing$, no point evaluation extends s_0: clause (i) holds. But Navara and Pták's m is a non-Dirac σ-additive two-valued state with $m \restriction B = s_0$; clause (ii) *fails*, so s_0 is not σ-essential. The construction *builds the rescuer*. This is exactly why clause (i) alone never suffices and the whole question is clause (ii): one must produce a carrier on which the point-level obstruction *cannot* be repaired by any non-Dirac state. Their carrier $\mathcal{L}_{f,g}$, being generated by two compatible (commuting) observables, is Boolean enough to admit the repair.

4 The bare statement

We now remove all lattice scaffolding from the σ-point-selection core and state it as a self-contained set-theoretic existence problem. A two-valued state on $\mathcal{L}$ selects, within each block (maximal compatible context, a countable partition of Ω into elements of $\mathcal{L}$), exactly one "true" cell; the selections across overlapping blocks must agree on shared events; σ-additivity constrains the selection over countable joins; non-Dirac means the selection is realised by no single point.

Question 4.1 (σ-point-selection, bare form)**.** Given a set Ω and a family $\{\Pi_\alpha\}$ of countable partitions of Ω (the blocks of a concrete σ-OML, pairwise overlapping in their common refinements), does there exist a selection assigning to each Π_α one cell $c_\alpha \in \Pi_\alpha$ such that

(a) (coherence) the selections agree on events common to several blocks;

(b) (σ-additivity) the induced $\{0, 1\}$-valuation commutes with countable disjoint unions;

(c) (non-principality) there is no $\omega_0 \in \Omega$ with $c_\alpha =$ the cell of Π_α containing ω_0 for all α;

(d) (prescription) the selection realises a prescribed finite pattern that no single point realises?

This is the object the Localization Theorem isolates: a witness for Definition 1.1 exists iff Question 4.1 has a positive answer for some admissible carrier. The bare form makes two features legible that the lattice language obscures.

Remark 4.2 (Why this is not simply a measurable cardinal)**.** If the blocks $\{\Pi_\alpha\}$ were *compatible* — a single Boolean base — a coherent non-principal σ-additive $\{0, 1\}$-selection would be exactly a non-principal countably-complete ultrafilter, hence a measurable cardinal [12]. The defining feature of Question 4.1 is that the blocks are *incompatible*: the coherence condition (a) is imposed across *overlapping, mutually non-distributive* contexts, and is strictly stronger than an ultrafilter condition on any single block. Consequently Question 4.1 does not reduce to the existence of a measurable cardinal; its Boolean shadow does, but the non-distributive coherence requirement is precisely what the reduction cannot cross. We make this non-reduction precise against the operator-algebra literature in §5.

Remark 4.3 (Relation to sheaf-theoretic contextuality)**.** Question 4.1 is the assertion that a system of locally coherent $\{0, 1\}$-valuations admits no *global section* of the corresponding presheaf — the sheaf-theoretic signature of contextuality in the sense of Abramsky and Brandenburger [2], in its σ-additive form [1]. The novelty here is not the sheaf framing, which is standard, but the conjunction of three features that the existing contextuality literature does not treat together: countable additivity, a non-Hilbert (abstractly concrete) carrier, and a $\{0, 1\}$-valued (non-probabilistic) selection. It is exactly this conjunction that is open.

5 The boundary map

We locate Question 4.1 by recording where its answer is known and where the obvious bridges fail.

The Polish boundary: empty above it. Derr and Williamson [6] prove a σ-extension criterion for coherent probabilities on pre-Dynkin systems: when the carrier is *Polish-representable* — Ω Polish, the generated σ-class equal to the Borel σ-algebra, with inner-regular blocks — finite coherence implies σ-extendability (their Theorem D.6, via Maharam [15]). Specialised to two-valued states, this says that on a Polish-representable carrier every finitely coherent s_0 globalises: *no σ-essential contextual state exists.* Hence a witness, if one exists, must live on a carrier that is *not* Polish-representable. This is the upper boundary of the problem.

The Hilbert boundary: the analogue exists, but does not transfer. Blecher and Weaver [5] prove that a singular countably additive *pure* state on the projection lattice of $B(\ell^2(\kappa))$ exists iff κ is Ulam measurable — a genuine large-cardinal-sensitive state-existence dichotomy on a non-distributive σ-complete OML. This is the nearest positive precedent: it shows that countable additivity of states on non-distributive structures *can* be set-theoretically sensitive. But it does not settle Question 4.1, for two reasons, neither of which is a reduction we can complete and both of which point the same way.

First, the carrier is of the wrong kind. The projection lattice of $B(H)$ for $\dim H \geq 3$ admits *no* two-valued state at all (Kochen–Specker [14]), so it is not concrete: it is not a sub-orthoposet of any powerset. A concrete σ-OML, by definition, has an order-determining family of two-valued states. The Blecher–Weaver object and ours are therefore not the same kind of structure, and their dichotomy concerns *pure* states (the relevant notion on $B(H)$), not two-valued ones, which are generically absent there by Gleason's theorem [9].

Second, the obvious transfer mechanism fails. The natural way to carry a state-existence result between a non-commutative algebra and an abelian (Boolean) subalgebra is to factor through a maximal abelian subalgebra. For the *pure*, two-valued case this factorisation does not hold: Akemann and Weaver [3] exhibit a pure state on $B(H)$ multiplicative on no masa, and Blecher–Weaver themselves rely on Marcus–Spielman–Srivastava paving precisely because the pure state does not reduce to a masa. So one cannot obtain a positive answer to Question 4.1 by importing Blecher–Weaver through a Boolean subobject.

Remark 5.1. Remarks 4.2 and the two paragraphs above are deliberately stated at the strength of *cited facts assembled*, not of a non-reducibility theorem. We do not claim a formal separation in any fixed reduction calculus; we claim, and the cited results establish, that the two obvious bridges — "it is a measurable cardinal" and "it transfers from Blecher–Weaver through a masa" — do not close Question 4.1, and that the obstruction in both cases is the same non-distributivity. This is what leaves the question open rather than a corollary of known work.

Below: faithful representation forces distributivity. One might hope to route Question 4.1 through a faithful σ-tribe representation — embedding $\mathcal{L}$ into a σ-algebra of honest sets so that two-valued states become point evaluations. This is unavailable: a faithful tribe-of-sets representation forces distributivity, hence Booleanness (the "Floor"; cf. [17]). The σ-Loomis–Sikorski theorem holds for structures with the Riesz Decomposition Property — σ-complete MV-algebras and RDP effect algebras [7] — but OMLs lack RDP (MO_2 is the standard witness), and the available representation is only a σ-epimorphic image that does not separate two-valued points. So the realization route that would trivialise Question 4.1 is blocked, for the same non-distributive reason.

Remark 5.2 (Segregated carriers do not witness)**.** A carrier is *segregated* when its contextuality is carried by a central Boolean factor — e.g. MO_ω or the product $\prod_n MO_2$, whose σ-additive two-valued states all concentrate at points. On such carriers every finitely coherent s_0 is the restriction of a Dirac, so clause (ii) of Theorem 3.2 fails and there is no witness. A witness must be *non-segregated*: the incompatibility must be irreducible, not routed through a central partition. This is why Definition 1.1 requires irreducibility, and it sharpens the search for a carrier to the non-segregated, non-Boolean, non-Polish-representable regime.

6 The open problem

Assembling §3–§5: the existence of a σ-essential contextual state is equivalent to a positive answer to Question 4.1, on a carrier that is concrete, σ-complete, non-Boolean, irreducible, non-segregated, and non-Polish-representable. The answer is not settled in either direction. It is not a corollary of the Blecher–Weaver dichotomy (the carrier is the wrong kind and the masa bridge fails), it does not reduce to a measurable cardinal (the non-distributive coherence condition is strictly stronger than its Boolean shadow), and it is not trivialised by a faithful representation (which would force distributivity).

Question 6.1 (Main)**.** Does there exist, in ZFC or under a large-cardinal hypothesis, a concrete σ-complete non-Boolean irreducible OML carrying a σ-essential contextual state? Equivalently, is the existence of a non-principal coherent σ-additive $\{0, 1\}$-selection across overlapping countable partitions (Question 4.1, non-segregated case) provable, refutable, or independent?

We record what a resolution would require. A positive ZFC answer would be a concrete construction crossing the non-Polish, non-segregated regime — which the present obstructions (Remarks 4.2, 5.1) show cannot be assembled from countably many compatible pieces, so it must be globally non-compositional. A negative answer would extend the Polish-representable emptiness of Derr–Williamson past its topological hypotheses to all concrete σ-OMLs, against the Blecher–Weaver evidence that the analogous Hilbert cell is inhabited under a large cardinal. An independence result would supply a model in which the selection exists and one in which it does not; the precedents for such results in the set theory of measure-theoretic structures (the von Neumann–Maharam problem [4], real-valued measurable cardinals [8]) are all real-valued and strictly positive, leaving the two-valued, non-distributive case — Question 6.1 — as the open edge.

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References

[1] Samson Abramsky and Rui Soares Barbosa. The sheaf-theoretic structure of definite causality. *Communications in Mathematical Physics*, 391, 2022. σ-additive sheaf-theoretic contextuality; see also arXiv:1905.08267.

[2] Samson Abramsky and Adam Brandenburger. The sheaf-theoretic structure of non-locality and contextuality. *New Journal of Physics*, 13:113036, 2011.

[3] Charles A. Akemann and Nik Weaver. $\mathcal{B}(H)$ has a pure state that is not multiplicative on any masa. *Proceedings of the National Academy of Sciences*, 105(14):5313–5314, 2008.

[4] Bohuslav Balcar, Thomas Jech, and Tomáš Pazák. Complete ccc Boolean algebras, the order sequential topology, and a problem of von Neumann. *Bulletin of the London Mathematical Society*, 37(6):885–898, 2005.

[5] David P. Blecher and Nik Weaver. Quantum measurable cardinals. *Journal of Functional Analysis*, 273(5):1870–1890, 2017.

[6] Rabanus Derr and Robert C. Williamson. Systems of precision: coherent probabilities on pre-Dynkin-systems and coherent previsions on linear subspaces. *arXiv preprint arXiv:2302.03522*, 2023.

[7] Anatolij Dvurečenskij. Loomis–sikorski theorem for σ-complete MV-algebras and ℓ-groups. *Journal of the Australian Mathematical Society*, 68(2):261–277, 2000.

[8] David H. Fremlin. *Measure Theory, Volume 5: Set-theoretic Measure Theory.* Torres Fremlin, 2008.

[9] Andrew M. Gleason. Measures on the closed subspaces of a Hilbert space. *Journal of Mathematics and Mechanics*, 6(6):885–893, 1957.

[10] Stanley P. Gudder. Quantum probability spaces. *Proceedings of the American Mathematical Society*, 21:296–302, 1969.

[11] John Harding. Remarks on concrete orthomodular lattices. *International Journal of Theoretical Physics*, 43(10):2149–2168, 2004.

[12] Thomas Jech. *Set Theory.* Springer, 3rd millennium edition, 2003.

[13] Gudrun Kalmbach. *Orthomodular Lattices.* Academic Press, 1983.

[14] Simon Kochen and Ernst P. Specker. The problem of hidden variables in quantum mechanics. *Journal of Mathematics and Mechanics*, 17(1):59–87, 1967.

[15] Dorothy Maharam. Consistent extensions of linear functionals and of probability measures. In *Proceedings of the Sixth Berkeley Symposium on Mathematical Statistics and Probability, Vol. II: Probability Theory*, pages 127–147. University of California Press, Berkeley, 1972.

[16] Mirko Navara and Pavel Pták. Two-valued measures on σ-classes. *Časopis pro pěstování matematiky*, 108(3):225–229, 1983.

[17] Pavel Pták and Sylvia Pulmannová. *Orthomodular Structures as Quantum Logics*, volume 44 of *Fundamental Theories of Physics.* Kluwer, Dordrecht, 1991.

[18] Roman Sikorski. *Boolean Algebras.* Springer, 2nd edition, 1964.

[19] Ron Wright. The state of the pentagon: a nonclassical example. In A. R. Marlow, editor, *Mathematical Foundations of Quantum Theory*, pages 255–274. Academic Press, New York, 1978.